

Assignment 53

1. (a) $G'(s) = -24.58751 \text{ tons/hour}^2$

(b) Let $A(t)$ be total amount of gravel in tons at time t

$$A(t) = A(0) + \int_0^t G(x) dx$$

$$A(8) = 500 + \int_0^8 G(t) dt = 1325.55109 \text{ tons}$$

(c) $A'(t) = G(t)$

$$A'(5) = G(5) = 98.14076$$

Increasing because $A'(5) = G(5) > 0$

(d) $A'(t) = G(t) = 0 \Rightarrow t =$

2. (a) $f(x) = g(x) \Rightarrow x = 0, 1.032832, 2, 3.28209$

$\sim g(x)$ Area_R = $\int_0^{1.032832} g(x) - f(x) dx = 0.99743$

$\sim f(x)$ Area_S = $\int_{1.032832}^2 f(x) - g(x) dx = 1.00692$

(b) Volume = $\int_{1.032832}^2 (f(x) - g(x))^2 dx = 1.28316$

(c) $h = f(x) - g(x)$

$h'(1.8) = -3.81172$

$$3. (a) \frac{v(20) - v(12)}{20 - 12} = \frac{240 - 200}{8} = \boxed{5}$$

$$\frac{16}{25} = \frac{80}{100}$$

~~$$f(x): y - 200 = 5(x - 12) \Rightarrow y = 5x + 140$$~~
~~$$y = 5(16) + 140 = \boxed{220}$$~~

(b) $\int_0^{40} |v(t)|$ is the total distance ~~traveled~~
 Johann traveled on her/his jog.

$$RPR = \sum_{i=0}^{n-1} f(x_{i+1}) (x_{i+1} - x_i)$$

$$\begin{aligned} & \downarrow \\ & 200(12) + 240(8) + (-220)(4) + 150(16) \\ & = 2400 + 1920 + 880 + 2400 = 4320 + 20 = \boxed{4340 \text{ meters}} \end{aligned}$$

$$(c) B'(t) = 3t^2 - 12t$$

$$B'(5) = 75 - 60 = \boxed{15 \text{ m/min}^2}$$

$$(d) \frac{1}{b-a} \int_a^b f(x) dx$$

$$\frac{1}{10} \int_0^{10} t^3 - 6t^2 + 300 = \frac{1}{10} \left(\frac{t^4}{4} \Big|_0^{10} - 2t^3 \Big|_0^{10} + 300t \Big|_0^{10} \right)$$

$$= \frac{1}{10} \left(\frac{10000}{4} - 2000 + 3000 \right) = \frac{1}{10} (2500 - 2000 + 3000)$$

$$= \frac{1}{10} (3500) = 350 \text{ meters/minute}$$